Major Project Proposal: MCA(IV), 2026

Project Title: **EventFlow: Seamless End-to-End Event Management & Ticketing**

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Learning Outcomes

- **Integrate Spring Security** with OAuth2 and OpenID Connect for secure, role-based access control.
- **Design and implement** a robust RESTful API architecture using Spring Boot.
- **Model complex domain relationships** and cardinalities (Events, Ticket Types, QR Validations).
- **Develop a dynamic frontend** using React and NPM to handle multi-user flows.
- **Implement real-time inventory management** logic to prevent ticket overselling.
- **Generate and validate unique QR codes** for secure physical entry management.

Description

A comprehensive full-stack event management platform that facilitates the entire lifecycle of an event. It enables organizers to create events with tiered pricing, allows attendees to discover and purchase tickets via a secure mobile-friendly interface, and provides staff with a specialized scanning module to validate entries in real-time.

Goals

- Provide a unified ecosystem for event creation, sales, and on-site validation.
- Ensure data integrity and security across three distinct user personas (Organizer, Attendee, Staff).
- Offer real-time sales analytics and attendance reporting for organizers.

Objectives

- **Onboarding:** Build a dashboard for organizers to manage multiple events and ticket tiers.
- **Discovery:** Implement a searchable event portal with a secure checkout flow.
- **Security:** Generate unique, status-tracked QR codes for every purchased ticket.
- **Validation:** Develop a high-speed mobile validation interface for event entry staff.
- **Analytics:** Create automated reporting systems for sales metrics and entry statistics.

Issues & Challenges

- **Concurrency:** Managing ticket inventory during high-traffic "entry rushes" to avoid overselling.
- **Security:** Protecting against ticket duplication and ensuring QR codes cannot be reused once scanned.
- **Environment:** Ensuring the mobile scanning interface remains performant in low-light or high-crowd venue conditions.
- **State Management:** Maintaining synchronization between the Spring Boot backend and React frontend during complex multi-step purchases.

Platform Tools & Methods

- **Backend:** Java 21, Spring Boot 3.x, Spring Data JPA, PostgreSQL.
- **Security:** Spring Security, OAuth2, OpenID Connect.
- **Frontend:** React, Tailwind CSS, Axios, NPM.
- **Mobile:** Responsive Web Design (optimized for mobile QR scanning).
- **Methods:** Domain-Driven Design (DDD), RESTful API standards, QR Code generation (ZXing), and Noun-Verb Analysis.

Deliverables

- **Functional Backend:** API with documented REST endpoints.
- **Web Application:** Responsive React views for Organizers, Attendees, and Staff.
- **Validation Engine:** QR code generation and real-time scanning logic.
- **Database:** Schema and Migration scripts (Flyway/Liquibase).
- **Documentation:** API specifications, environment setup, and user manual.

Action Plan:

Week	Focus Area	Key Tasks
Week 1-2	**Design**	Domain analysis, System Design, and Database Schema implementation.
Week 3	**Setup**	Spring Boot initialization and Security (OAuth2) integration.
Week 4	**Core API**	Event creation and Management APIs (CRUD).
Week 5	**Sales Logic**	Purchase flow, payment simulation, and inventory control.
Week 6	**Validation**	QR code generation and Staff scanning interface.
Week 7	**Frontend**	Building React views and state management for all user flows.
Week 8	**Polishing**	Sales analytics dashboard, testing, and final deployment.

Signature: **Trainee:** **Mentor:**